

MCA Semester 1	Subject : Advanced Data Structures Lab
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Roll No. : MCA2511	Date : 01-12-2025

Code:

```
import java.util.Scanner;
```

```
class AdjMat
```

```
{
```

```
    public static void main(String[] args)
```

```
    {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        //No. of vertices
```

```
        System.out.print("Enter the number of vertices in the graph: ");
```

```
        int v = sc.nextInt();
```

```
        //Initialize adjacency matrix
```

```
        int[][] mat = new int[v][v];
```

```
        //No of edges
```

```
        System.out.print("Enter the number of edges: ");
```

```
        int e = sc.nextInt();
```

```
        //Read the edges
```

```
        for(int i = 1; i<=e; i++)
```

```
        {
```

```
            System.out.println("Edge "+ i +" source destination: ");
```

```
            int source = sc.nextInt();
```

```
            int dest = sc.nextInt();
```

```
            mat[source][dest] = 1;
```

```
            mat[dest][source] = 1; //Comment this line for a Directed Graph
```

```
        } //end of for
```

```
        //Display
```

```
        System.out.println("\nAdjacency Matrix: ");
```

```
        for(int i = 0; i<v; i++)
```

```
        {
```

```
            for(int j = 0; j<v; j++)
```

```
            {
```

```
                System.out.print(mat[i][j] + " ");
```

```
            } //end of for j
```

```
        System.out.println();  
    } //end of for i  
} //end of psvm  
} //end of AdjMat
```

Output:

```
A:\MCA2511\DS_LAB>javac 19AdjMat.java  
  
A:\MCA2511\DS_LAB>java AdjMat  
Enter the number of vertices in the graph: 5  
Enter the number of edges: 6  
Edge 1 source destination:  
0 1  
Edge 2 source destination:  
0 3  
Edge 3 source destination:  
1 2  
Edge 4 source destination:  
1 3  
Edge 5 source destination:  
2 4  
Edge 6 source destination:  
3 4  
  
Adjacency Matrix:  
0 1 0 1 0  
1 0 1 1 0  
0 1 0 0 1  
1 1 0 0 1  
0 0 1 1 0
```

```
A:\MCA2511\DS_LAB>java AdjMat
Enter the number of vertices in the graph: 6
Enter the number of edges: 7
Edge 1 source destination:
0 1
Edge 2 source destination:
1 2
Edge 3 source destination:
1 4
Edge 4 source destination:
2 4
Edge 5 source destination:
2 3
Edge 6 source destination:
4 3
Edge 7 source destination:
4 5
```

```
Adjacency Matrix:
0 1 0 0 0 0
1 0 1 0 1 0
0 1 0 1 1 0
0 0 1 0 1 0
0 1 1 1 0 1
0 0 0 0 1 0
```